Anjali Devi Sivakumar

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Education

University of Michigan, Ann Arbor, MI *Sept 2022 - Dec 2025(anticipated)*
Ph.D. in Electrical and Computer Engineering, GPA: 4.0 /4.0
Advisors: Prof. Xudong (Sherman) Fan and Prof. Zhaohui Zhong

University of Michigan, Ann Arbor, MI *Jan 2021- May 2023*
Master of Science in Electrical and Computer Engineering, GPA: 4.0 /4.0

SASTRA Deemed to be University, Tamil Nadu, India *Jun 2015 - Jun 2019*
Bachelor of Technology in Electronics and Instrumentation Engineering, GPA: 9.31/10.0

Research Experience

Graduate Student Research Assistant *Jan 2021- Present*
University of Michigan, Ann Arbor, MI
Advisors: Prof. Xudong (Sherman) Fan and Prof. Zhaohui Zhong

- Designed and characterization of a microfluidic Differential Photoionization Detector (D-PID) for ultra-trace analyte detection, enhancing resolution in MEMS-based wearable micro-GC systems.
- Developed a low-power, low-cost 1D Gas Chromatography (GC) system for disease diagnosis through body odor analysis using MEMS-based micro-GC.
- Designed the scalable Wearable Analytical Skin Probe (WASP) for monitoring skin dynamics and tracking atopic skin diseases in collaboration with Michigan Medicine.
- Contributed to developing portable 1D GC breath analyzers for non-invasive COVID-19 detection via VOC analysis, leveraging MEMS sensors in collaboration with Michigan Medicine.

Research Assistant *Jan 2019 - Nov 2020*
Harvard Medical School – Brigham and Women's Hospital, Cambridge, MA
Advisor: Prof. Hadi Shafiee

- Assisted in designing and testing microfluidic-based, deep learning-powered self-testing diagnostic devices for rapid, accurate point-of-care detection of emerging infectious pathogens (COVID-19, HIV, HCV, HBV).
- Contributed to the development of portable, cost-effective smartphone-based diagnostic systems for in vitro fertilization (IVF) applications.
- Collaborated with Harvard School of Public Health to investigate the effects of environmental exposures on semen quality and sperm epigenome using a smartphone-based diagnostic device.
- Conducted histological analysis of cells to train machine-learning models for accurate morphological classification at varying image resolutions.
- Performed psychophysical studies on human skin perception to prototype a haptic glove aimed at enhancing tactile feedback.

Skills

- **Programming Languages:** C, C++, Python, Embedded C, MATLAB
- **Simulation Software**: LT-Spice, Tina, Simulink, Cadence Virtuoso, LabVIEW, COMSOL
- **Rapid Prototyping Hardware:** Laser cutting, 3-D Printing
- **Operating Systems:** Windows, Linux (Ubuntu), Mac OSX
- **Robot Platforms:** ROS
- **Microfabrication**
- **Design Software:** AutoCAD fusion, KiCAD, Altium, Klayout
- **Wet Lab:** Histology, Cell culture, Biosafety Level 2+

Selected Journal Publications

- **Anjali Devi Sivakumar**, Ruchi Sharma, Chandrakalavathi Thota, Ding Ding, and Xudong Fan, "WASP: Wearable Analytical Skin Probe for Dynamic Monitoring of Transepidermal Water Loss", ***ACS Sensors* 8** , 4407-4416, November 2023**.**
- Xiaheng Huang, Ruchi Sharma, **Anjali Devi Sivakumar**, Shuo Yang, and Xudong Fan, "Ultrathin silica integration for enhancing reliability of microfluidic photoionization detectors", ***Analytical Chemistry 95***, 8496–8504, May 2023.
- Ruchi Sharma, Wenzhe Zang, Ali Tabartehfarahani, Andres Lam, Xiaheng Huang, **Anjali Devi Sivakumar**, et al., "Portable Breath-Based Volatile Organic Compound Monitoring for the Detection of COVID-19 During the Circulation of the SARS-CoV-2 Delta Variant and the Transition to the SARS-CoV-2 Omicron Variant", ***JAMA Network Open 6***, e230982, February 2023.
- Xiaheng Huang, Maxwell Wei-hao Li, Wenzhe Zang, Xiaolu Huang, **Anjali Devi Sivakumar**, Ruchi Sharma, and Xudong Fan, "Portable Comprehensive Two Dimensional Micro-Gas Chromatography Using Integrated Flow-Restricted Pneumatic Modulator", ***Nature Microsystems & Nanoengineering* 8**, 115, November 2022.
- Manoj Kumar Kanakasabapathy, Prudhvi Thirumalaraju, Hemanth Kandula, Fenil Doshi, **Anjali Devi Sivakumar,** et al., "Adaptive Adversarial Neural Networks for Lossy and Domain-Shifted Medical Image Analysis", ***Nature Biomedical Engineering* 5**, 571-585, June 2021.

Selected Patents

- Xudong Fan, Xiaheng Huang, Shuo Yang, Wencheng Li, Robert Nidetz, Ruchi Sharma, **Anjali Devi Sivakumar**, et al., 'Monolithic integration of micro-gas chromatography column and micro-photoionization detector on a chip', (Provisional patent application number: 63/622,741, filed on January 19, 2024).
- Xudong Fan, **Anjali Devi Sivakumar**, 'A transepidermal water loss device' (PCT application number: 18/428,514, filed on February 1, 2024).
- Xiaheng Huang, Ruchi Sharma, **Anjali Devi Sivakumar**, Shuo Yang, and Xudong Fan, 'Protective Dielectric Layers for Enhancing Reliability of Vacuum Ultraviolet Lamps' (PCT application number: PCT/US2024/022212, filed on March 29, 2024).

Selected Academic Honors and Awards

- Received the **Dr. Helen Wu Award** from Rackham Graduate School for Spring/Summer 2024.
- Awarded the **Barbour Scholarship** from Rackham Graduate School for the academic year 2024-25.
- Received the **Rackham Conference Travel Grant** from Rackham Graduate School for the academic year 2024-25.
- Inducted into the **Excellence in ECE Honor Roll 2023-24** for service to the ECE department through contributions to department-specific events and programming designed to foster an inclusive, supportive community for all.
- Awarded **the Research Excellence Award** from **Brigham Research Institute** for **outstanding contributions to research** at **Brigham and Women's Hospital** on Nov-12,2020**.**
- Received the **Desh Videsh Scholarship** for pursuing my bachelor's thesis at Harvard Medical School, MA, USA.
- Awarded the **Smt. K.Thulasi Memorial Award** for **best outgoing student from School of Electrical and Electronics Engineering** amongst the graduating batch of 2019.
- Received the **Dean's Merit Scholarship** for being in the **top 2% of UG Batch 2015-2018** in terms of academic performance.

Professional and Service Activities

- **Member**, Lurie Nano Fabrication (LNF) User Committee, University of Michigan
- ***Research Mentor,*** Undergraduate Research Opportunity Program (UROP), University of Michigan
- ***K12 Outreach Officer***, Graduate Society of Women Engineers (GradSWE), University of Michigan
- ***Active Member***, Tau Beta Pi (TBP) - The Engineering Honor Society, University of Michigan
- ***Active Member***, Eta Kappa Nu (HKN) – The Electrical and Computer Engineering Honor Society, University of Michigan